9.6: Method of Moments Estimation

Recall:
- For a random variable Y, the k-th **raw moment** is $\mu'_k = E\left[Y^k\right]$.
- The k-th **sample raw moment**, based on a random sample of size n is $m'_k = \frac{1}{n}\sum_{i=1}^{n} Y_i^k$.

Method of Moments Estimation:

Choose as estimates for $\theta = (\theta_1, \theta_2, \ldots, \theta_t)$, the values that are the solutions to

$$\mu'_k = m'_k, \quad k = 1, 2, \ldots .$$

Let $\hat{\theta}_{MM}$ denote the Method of Moments Estimator for θ.

Find the Method of Moments Estimators

Example 3: $X_1, \ldots, X_n \sim Bernoulli(p)$.

$P(X=1)=p$

$P(X=0)=1-p$

$E[X] = p \approx \bar{Y}, \hat{p}_{mm} = \bar{Y}$

Example 4: $Y_1, \ldots, Y_n \overset{ind.}{\sim} Poisson(\lambda)$. iid

$E[Y] = \lambda \approx \bar{Y}, \quad \hat{\lambda}_{mm} = \bar{Y}.$

Example 6: $Y_1, \ldots, Y_n \overset{ind.}{\sim} Exponential(\theta)$.

$E[Y] = \theta, \quad \hat{\theta}_{mm} = \bar{Y}$

Example 16: $Y_1, \ldots, Y_n \overset{ind.}{\sim} Geometric(p)$.

$E[Y] = \frac{1}{p}, \quad p = \frac{1}{E[Y]} \approx \frac{1}{\bar{Y}}, \quad \frac{1}{\bar{Y}}$ is mme

Example 5: $X_1, \ldots, X_n \overset{ind.}{\sim} Uniform(0, \theta)$.

$E[Y] = \frac{\theta}{2}, \quad \theta = 2E[Y] \approx 2\bar{Y}$

Example 10: $Y_1,\ldots,Y_n \overset{ind.}{\sim} Beta(\alpha,\alpha).$

$$\text{Beta}(\alpha,\beta) = \begin{cases} \dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, & x \in (0,1) \\ 0, & \text{otherwise} \end{cases}$$

$$E\{Y\} = \frac{\alpha}{\alpha+\beta} \qquad E\{Y^2\} = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} + \frac{\alpha^2}{(\alpha+\beta)^2}$$

in the case $\alpha=\beta$

$$E\{x\} = \frac{1}{2}$$

$$E\{x^2\} = \frac{\alpha^2}{4\alpha^2(2\alpha+1)} + \frac{1}{4} = \frac{1}{4(2\alpha+1)} + \frac{1}{4}$$

$$4E\{x^2\} = 1 + \frac{1}{2\alpha+1}$$

$$2\alpha+1 = \frac{1}{4E\{x^2\}-1}, \quad \alpha = \frac{1-2E\{x^2\}}{4E\{x^2\}-1}$$

$$\alpha \approx \frac{1 - 2\overrightarrow{Y^2}}{4\overrightarrow{Y^2} - 1}$$

$$Y \sim N(\mu_Y, \sigma_Y^2)$$

$$\mu_Y = E[Y] \approx \overline{Y}$$

$$\sigma_Y^2 = \underbrace{E[Y^2]}_{\frac{22}{Y^2}} - \underbrace{(E[Y])^2}_{(\overline{Y})^2}$$

$$\sigma_{mm}^2 = \overline{Y^2} - (\overline{Y})^2$$

$$\mu_{mm} = \overline{Y}$$

Example 11: $Y_1, \ldots, Y_n \overset{ind.}{\sim} Gamma(\alpha, \beta)$.

$$f_Y(y) \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} y^{\alpha-1} e^{-y/\beta}, & y > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\mu_Y = E\{Y\} \approx \bar{Y}$$

$$\sigma_Y^2 = \underbrace{E\{Y^2\}}_{\frac{\Sigma Y^2}{n}} - \underbrace{(E\{Y\})^2}_{(\bar{Y})^2}$$

$$\sigma^2_{mm} = \overline{Y^2} - (\bar{Y})^2$$

$$\mu_{mm} = \bar{Y}$$

Example:

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed random variables with mean μ_Y and variance σ_Y^2. Find the method of moments estimators for μ_Y and σ_Y^2.